DATA 260 Homework 8 - Kafka Restaurant Inspection Service

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Configuration: PORT_BASE=8085, PREFIX=s7085, SEED=7085, VERIFY_SEED=267085, DOMAIN_ID=5. Hardware/model: as documented in HW7; local model llama3.2:3b. Final commit/tag is recorded in verification.json after packaging.

Section 0 - Scope and evidence

This report extends the restaurant-inspection application from HW7. The implementation includes an aiokafka runtime plus deterministic offline tests. Live Docker/Kafka evidence is only claimed when the command actually ran; offline evidence is labeled as such.

Part 1 - Request/response service

Topics: s7085_entity_requests and s7085_entity_responses. Requests contain correlation_id, reply_to, operation, and data. CREATE_ENTITY validates restaurant inspection fields, generates a UUID, and stores the record in the worker dictionary. GET_ENTITY returns the record or a clean not-found error. Correlation IDs are echoed on all responses.

Relevant code: kafka_protocol.py, kafka_runtime.py, kafka_rpc.py, routers/hw8.py, homework_demo.py.

Part 2 - Three-agent pipeline

Planner, Writer, and Reviewer are separate entry points. The real runtime uses Kafka consumer/producer stages and manual offset commits. The deterministic pipeline records the same four-topic journey for repeatable smoke tests: s7085_inbox -> s7085_tasks -> s7085_drafts -> s7085_final.

Part 3 - Failure behavior and ordering

Scenario	Before	After
Duplicate	Replay could repeat side effect	correlation-id cache returns the original response
Consumer death	uncommitted work can redeliver	commit occurs after successful publish
Poison	bad message can block	bounded retries publish s7085_dead_letter
Ordering	two partitions may interleave	entity key keeps related records in one partition

The raw artifact part3_failure_ordering.json records VERIFY_SEED=267085, 20 messages, duplicate idempotency, consumer restart/commit semantics, retry count, and dead-letter result. Kafka provides decoupling and independent scaling; it adds responsibilities for idempotency, retries, offsets, and partition-key ordering.

Verification and limitations

Run python verify_hw8.py to write reports/hw08/verification.json. The test suite covers validation, correlation preservation, UUID round trips, duplicate delivery, pipeline output, dead-letter configuration, and keyed ordering. Live Docker/Kafka execution, JMeter benchmarking, and collaborator-permission verification remain environment/account checks unless their logs are present in RUN_LOG.txt.

AI_USE

AI assistance was used for scaffolding, review, and report formatting. I selected the restaurant domain, supplied configuration, inspected the outputs, and verified tests. I independently checked correlation preservation and the commit/tag sequencing. Changes were retained only when the source, deterministic

tests, and artifacts agreed.